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Computational Logic Handouts

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Chapter 1

LoDE

Exercise 1.1 Considering the following TBox and ABoxes, determine which of the statements below are true. Assume that "Anon#1" is a placeholder for any anonymous entity added during the expansion of an ABox.

$City \sqsubseteq \exists name.String$
 $Person \sqsubseteq \exists bornIn.City$
 $Book \sqsubseteq \exists title.String \sqcap \exists topic.String$
 $Philosophy \sqsubseteq \exists mainBelief.String \sqcap Discipline$
 $Citizen \equiv Person \sqcap \exists address.String \sqcap \exists livesIn.City$
 $Philosopher \equiv Person \sqcap \exists expertIn.Philosophy \sqcap \forall wrote.Book$

ABox1

$Citizen(Bill\#1)$
 $Person(Sara\#1)$
 $City(Trento\#1)$

ABox2

$Philosopher(Eva\#1)$
 $Philosophy(Ph\#1)$
 $wrote(Eva\#1, Bk\#1)$

ABox3

$Philosopher(Eva\#1)$
 $City(Trento\#1)$
 $Book(Bk\#1)$

Language and Knowledge reasoning

1. $T \models Citizen \sqperp Philosopher$.
2. $Discipline$ is a primitive etype.
3. The definition of $Philosopher$ is the only one in the theory that can be unfolded.

Entity reasoning

1. Considering CWA, we know that $T \cup ABox1 \models livesIn(Bill\#1, Trento\#1)$ is false.
2. By completely expanding ABox2 we add 4 anonymous entities of type $String$.
3. Considering OWA, it's possible to know if $T \cup ABox2 \models expertIn(Eva\#1, Ph\#1)$ is true.
4. Considering OWA, $T \cup ABox2 \models Book(Bk\#1)$
5. Considering CWA, $T \cup ABox3 \models wrote(Eva\#1, Bk\#1)$

Exercise 1.2 Considering the following TBox and ABoxes, determine which of the statements below are true. Assume that "Anon#1" is a placeholder for any anonymous entity added during the expansion of an ABox.

$RestEmployee \equiv Person \sqcap \forall worksAt. Restaurant$
 $RestOwner \equiv Person \sqcap \forall ownsA. Restaurant$
 $Cook \equiv RestaurantEmployee \sqcap \exists canPrepare. Recipe$
 $Waiter \equiv RestaurantEmployee \sqcap \exists serve. Table$
 $Chef \equiv Cook \sqcap \exists headOf. Kitchen$
 $Waiter \sqsubset Cook$
 $RestaurantEmployee \sqsubset RestaurantOwner$

ABox1	ABox2	ABox3
$Chef(Mike\#1)$	$Chef(Mike\#1)$	$Cook(Mike\#1)$
$Waiter(Will\#1)$	$RestEmployee(Will\#1)$	$Waiter(Will\#1)$
$RestOwner(Robert\#1)$	$RestOwner(Robert\#1)$	$canPrepare(Mike\#1, R\#1)$
	$ownsA(Robert\#1, R\#1)$	

Language and Knowledge reasoning

1. $T \models Waiter \sqsubseteq \neg RestOwner$
2. $T \models Cook \sqsubseteq \neg \exists ownsA. \neg Restaurant$
3. $T \models Chef \sqcap Waiter \equiv \perp$

Entity reasoning

1. Considering CWA, we know that $T \cup ABox1 \models worksAt(Mike\#1, Anon\#1)$
2. The expansion of $Chef(Mike\#1)$ in ABox1 generates two new entities to be added to the ABox
3. The expansion of $RestOwner(Robert\#1)$ in ABox2 generates one new entity to be added to the ABox.
4. Considering CWA, in ABox3, $R\#1$ is a *Recipe*.

Exercise 1.3 Considering the following TBox and ABoxes, determine which of the statements below are true. Assume that "Anon#1" is a placeholder for any anonymous entity added during the expansion of an ABox.

$Employee \equiv Person \sqcap \exists worksAt.ShopPlace$

$ShopPlace \equiv Place \sqcap \exists sells.Thing$

$Bartender \equiv Employee \sqcap \forall worksAt.Bar$

$Bar \equiv ShopPlace \sqcap \forall sells.Drink$

ABox1

$Employee(E\#1)$

$ShopPlace(S\#1)$

$Thing(T\#1)$

ABox2

$Bartender(E\#1)$

$Bar(S\#1)$

$Drink(T\#1)$

ABox3

$Bartender(E\#1)$

$worksAt(E\#1, S\#1)$

$sells(S\#1, T\#1)$

Language and Knowledge reasoning

1. $T \models Bar \sqsubseteq \neg \forall sells. \neg Thing$
2. $T \models ShopPlace \sqsubseteq Place$
3. $T \models Bartender \sqsubseteq \exists worksAt.Bar$

Entity reasoning

1. Considering OWA, $T \cup ABox3 \models Drink(T\#1)$ is true
2. By expanding ABox2, we add the assertion $worksAt(E\#1, Anon\#1)$
3. Considering CWA, $T \cup ABox1 \models sells(S\#1, T\#1)$
4. Considering CWA, $T \cup ABox2 \models sells(S\#1, Anon\#1)$ and $T \cup ABox2 \models Drink(Anon\#1)$
5. Considering OWA, by expanding ABox1, we know that the *ShopPlace* *S#1* and the *ShopPlace* *Anon#1* in which *E#1* works, sell the same *Thing* (the same entity of type *Thing*).
6. Considering CWA, $T \cup ABox3 \models Thing(T\#1)$

1.1 Solutions

Solution 1.1

Unfolded TBox

City $\sqsubseteq \exists \text{name}.String$

Person $\sqsubseteq \exists \text{bornIn}.(\exists \text{name}.String)$

Book $\sqsubseteq \exists \text{title}.String \sqcap \exists \text{topic}.String$

Philosophy $\sqsubseteq \exists \text{mainBelief}.String \sqcap \text{Discipline}$

Citizen $\equiv \exists \text{bornIn}.(\exists \text{name}.String) \sqcap \exists \text{address}.String \sqcap \exists \text{livesIn}.(\exists \text{name}.String)$

Philosopher $\equiv \exists \text{bornIn}.(\exists \text{name}.String) \sqcap \exists \text{expertIn}.(\exists \text{mainBelief}.String \sqcap \text{Discipline})$
 $\sqcap \forall \text{wrote}.(\exists \text{title}.String \sqcap \exists \text{topic}.String)$

Expanded ABox1

Citizen(Bill#1), *bornIn*(Bill#1, C#1), *City*(C#1), *name*(C#1, S#1), *String*(S#1), *address*(Bill#1, S#2),
String(S#2), *livesIn*(Bill#1, C#2), *City*(C#2), *name*(C#2, S#3), *String*(S#3)
Person(Sara#1), *bornIn*(Sara#1, C#3), *City*(C#3), *name*(C#3, S#4), *String*(S#4)
City(Trento#1), *name*(Trento#1, S#5), *String*(S#5)

Expanded ABox2

Philosopher(Eva#1) *bornIn*(Eva#1, C#1), *City*(C#1), *name*(C#1, S#1), *String*(S#1),
expertIn(Eva#1, P#1), *Philosophy*(P#1), *mainBelief*(P#1, S#2), *String*(S#2), *Discipline*(P#1),
Book(Bk#1), *title*(Bk#1, S#3), *String*(S#3), *topic*(Bk#1, S#4), *String*(S#4)
Philosophy(Ph#1), *mainBelief*(Ph#1, S#5), *String*(S#5), *Discipline*(Ph#1)
wrote(Eva#1, Bk#1)

Expanded ABox3

Philosopher(Eva#1) *bornIn*(Eva#1, C#1), *City*(C#1), *name*(C#1, S#1), *String*(S#1),
expertIn(Eva#1, P#1), *Philosophy*(P#1), *mainBelief*(P#1, S#2), *String*(S#2), *Discipline*(P#1)
City(Trento#1), *name*(Trento#1, S#3), *String*(S#3)
Book(Bk#1), *title*(Bk#1, S#4), *String*(S#4), *topic*(Bk#1, S#5), *String*(S#5)

Language and Knowledge reasoning

1. false
2. true
3. false

Entity reasoning

1. true
2. false, it's 5
3. false
4. true
5. false

Solution 1.2**Unfolded TBox**

$RestEmployee \equiv Person \sqcap \forall worksAt. Restaurant$

$RestOwner \equiv Person \sqcap \forall ownsA. Restaurant$

$Cook \equiv Person \sqcap \forall worksAt. Restaurant \sqcap \exists canPrepare. Recipe$

$Waiter \equiv Person \sqcap \forall worksAt. Restaurant \sqcap \exists serve. Table$

$Chef \equiv Person \sqcap \forall worksAt. Restaurant \sqcap \exists canPrepare. Recipe \sqcap \exists headOf. Kitchen$

$Waiter \sqperp Cook$

$RestaurantEmployee \sqperp RestaurantOwner$

Expanded ABox1

$Chef(Mike\#1), Cook(Mike\#1), headOf(Mike\#1, K\#1), Kitchen(K\#1), RestEmployee(Mike\#1)$
 $Person(Mike\#1)$

$Waiter(Will\#1), RestEmployee(Will\#1), serve(Will\#1, T\#1), Table(T\#1), Person(Will\#1)$
 $RestOwner(Robert\#1), Person(Robert\#1)$

Expanded ABox2

$Chef(Mike\#1), Cook(Mike\#1), headOf(Mike\#1, K\#1), Kitchen(K\#1), RestEmployee(Mike\#1)$
 $Person(Mike\#1)$

$RestEmployee(Will\#1), Person(Will\#1)$

$RestOwner(Robert\#1), Person(Robert\#1), Restaurant(R\#1)$
 $ownsA(Robert\#1, R\#1)$

Expanded ABox3

$Cook(Mike\#1), RestEmployee(Mike\#1), Person(Mike\#1), canPrepare(R\#2), Recipe(R\#2)$
 $Waiter(Will\#1), RestEmployee(Will\#1), serve(Will\#1, T\#1), Table(T\#1), Person(Will\#1)$
 $canPrepare(Mike\#1, R\#1)$

Language and Knowledge reasoning

1. true
2. false
3. true

Entity reasoning

1. false
2. true
3. false
4. false

Solution 1.3**Unfolded TBox**

$Employee \equiv Person \sqcap \exists worksAt.(Place \sqcap \exists sells.Thing)$

$ShopPlace \equiv Place \sqcap \exists sells.Thing$

$Bartender \equiv Person \sqcap \exists worksAt.(Place \sqcap \exists sells.Thing)$

$\sqcap \forall worksAt.(Place \sqcap \exists sells.Thing \sqcap \forall sells.Drink)$

$Bar \equiv Place \sqcap \exists sells.Thing \sqcap \forall sells.Drink$

Expanded ABox1

$Employee(E\#1), Person(E\#1), worksAt(E\#1, P\#1), ShopPlace(P\#1), Place(P\#1),$
 $sells(P\#1, T\#2), Thing(T\#2)$

$ShopPlace(S\#1), Place(S\#1), sells(S\#1, T\#3), Thing(T\#3)$

$Thing(T\#1)$

Expanded ABox2

$Bartender(E\#1), Employee(E\#1), Person(E\#1), worksAt(E\#1, P\#1), ShopPlace(P\#1), Place(P\#1),$
 $sells(P\#1, T\#2), Thing(T\#2), Bar(P\#1), Drink(T\#2)$

$Bar(S\#1), ShopPlace(S\#1), Place(S\#1), sells(S\#1, T\#3), Thing(T\#3) Drink(T\#3)$

$Drink(T\#1)$

Expanded ABox3

$Bartender(E\#1), Employee(E\#1), Person(E\#1), Bar(S\#1), ShopPlace(S\#1),$
 $Drink(T\#1), sells(S\#1, T\#2), Thing(T\#2), Drink(T\#2)$

$worksAt(E\#1, S\#1)$

$sells(S\#1, T\#1)$

Language and Knowledge reasoning

1. true
2. false
3. true

Entity reasoning

1. true
2. false
3. false
4. true
5. false
6. false

1.2 Unfolding and expanding a LoDE theory

The following procedure is used to solve *LoDE entity reasoning problems* by reducing them to *LoE reasoning problems* after expanding the ABox of the theory with respect to its TBox.

Consider a *PrimitiveEtype* as an Etype that appears only on the right of a definition, consider every other Etype a *ComposedEtype*.

Let T be a *definitional* TBox and let A be its ABox.

1. Choose an assertion in A in the form $ComposedEtype(e\#1)$.
2. Retrieve the definition of $ComposedEtype$, $C_1 \sqcap \dots \sqcap C_n$, from T and apply the following expansion rules to each conjunct C_i , where $1 \leq i \leq n$:
 - if C_i is in the form $Etype$, and $Etype(e\#1) \notin A$, then
 $A = A \cup \{Etype(e\#1)\}$
 - if C_i is in the form $\exists r.E$, and there is no entity $e\#2$ such that $\{r(e\#1, e\#2), E(e\#2)\} \subseteq A$, then
 $A = A \cup \{r(e\#1, e\#2), E(e\#2)\}$ where $e\#2$ is new in A .
 - if C_i is in the form $\forall r.E$, and A contains an assertion $r(e\#1, e\#2)$, then
 $A = A \cup \{E(e\#2)\}$.
3. Reiterate the procedure from step 1 until no other assertion would be added by expanding another etype.

example

Consider a TBox $T = \{Female \sqsubseteq Person \sqcap \exists name.String, Mother \equiv Female \sqcap \exists hasChild.Person\}$ and an ABox $A = \{Mother(Sara\#1)\}$.

Let us say that we want to know if $T \cup A \models Person(Sara\#1)$.

$$\begin{array}{c}
 T = \begin{cases} Female \sqsubseteq Person \sqcap \exists name.String \\ Mother \equiv Female \sqcap \exists hasChild.Person \end{cases} \\
 \\
 \begin{array}{ccccc}
 & & & & Mother(Sara\#1) \\
 & & & & Female(Sara\#1) \\
 & & & & Person(Sara\#1) \\
 Mother(Sara\#1) & \leftarrow & \Rightarrow & \begin{array}{l} Mother(Sara\#1) \\ Female(Sara\#1) \\ hasChild(Sara\#1, e\#1) \\ Person(e\#1) \end{array} & \leftarrow & \Rightarrow & \begin{array}{l} Mother(Sara\#1) \\ Female(Sara\#1) \\ Person(Sara\#1) \\ hasChild(Sara\#1, e\#1) \\ Person(e\#1) \\ name(Sara\#1, s\#1) \\ String(s\#1) \end{array}
 \end{array}
 \end{array}$$

From the resulting ABox A' it is clear that $T \cup A \models Person(Sara\#1)$ is satisfied as $Person(Sara\#1) \in A'$.